Personal details and date of CV

- **Surname:** Zakary
- **First name:** Ouail
- **ResearcherID:** OSI-7733-2025
- **ORCID:** 0000-0002-7793-3306

- **Date of the CV:** 30.5.2026

Degrees

- **08.12.2023:** Ph.D. in Physics, Dept. of Physics, *Le Mans Université*, Avenue Olivier Messiaen, 72085 Le Mans cedex 9, France.
- **30.06.2020:** M.Sc. in Physics, Dept. of Physics, *Le Mans Université*, Avenue Olivier Messiaen, 72085 Le Mans cedex 9, France.
- **30.06.2020:** M.Sc. in Physics, Dept. of Physics, *Hassan II University of Casablanca*, 19 Rue Tarik Bnou Ziad, Mers Sultan, 20100 Casablanca, Morocco.
- **30.06.2018:** B.Sc. in Physics, Dept. of Physics, *Hassan II University of Casablanca*, 19 Rue Tarik Bnou Ziad, Mers Sultan, 20100 Casablanca, Morocco.

Language Skills

- **English** and **French** (Proficient, C2)
- **Arabic** (Native language)

Current employment

- **05.02.2024–04.02.2027:** Postdoctoral Researcher, *NMR Research Unit*, *University of Oulu*, Finland (Website: https://cc.oulu.fi/ nmrwww/index.html). Developing machine learning methodologies for molecular and solid-state materials; co-supervising Ph.D. students and teaching computational physics and chemistry courses.
- **Stage II** of academic research career on the four-stage (I–IV) research career model of Finland.

Previous work experience

- **01.10.2020–08.12.2023:** Doctoral Researcher (PhD Student), Department of Physics, *Institut des Molécules et Matériaux du Mans* (IMMM), *Le Mans Université*, Le Mans, France. Research focus: computational modeling of disorder in transition metal oxyfluorides.
- **04.04.2022–08.04.2022:** Visiting PhD Student working with Dr. Vincent Sarou-Kanian at the CEMHTI lab, Orléans, France.
- **01.05.2020–31.08.2020:** Internship at the *Institut des Molécules et Matériaux du Mans* (IMMM), *Le Mans Université*, Le Mans, France.
- **01.06.2019–31.08.2019:** Internship at the *Laboratoire de Physique de la Matière Condensée* (LPMC), *Hassan II University of Casablanca*, Casablanca, Morocco.

Research funding and grants

- **2024:** 3300 € from *Otto A. Malm Foundation*, travel grant for the project "*Realistic Modeling of Large-Scale Porous Materials using Machine Learning-Assisted Molecular Dynamics*".
- **2020–2023:** Excellence Grant, 6000 € from the Ministry of Higher Education and Research of Morocco, awarded for academic merit during doctoral studies at *Le Mans Université*, France.

Research output

***Publications*:**

- **Total:** 6 published scientific articles, 1 article submitted for publication, and 9 in preparation.

- **10 most important publications** (corresponding authorship marked with *):

1. **Zakary, O.***; Yin, W.; Aryal, N.* Local Symmetry Breaking and Two-Stage Phase Transition in RuP Uncovered by a Fine-Tuned Atomistic Foundation Model. *ChemRxiv* 2026, 0330, *Submitted for publication*.
   [DOI: 10.26434/chemrxiv.15001387/v1] [Supporting Data] [Supporting Code]
2. Laurila, O.; Jacklin, T.; **Zakary, O.***; Lantto, P.* Machine Learning-Accelerated Path Integral Molecular Dynamics and ^{13}C NMR Simulations Unlock New Insights Into Quantum Effects in C_{60} Fullerene. *J. Phys. Chem. A* **2026**, 130, 2169–2181.
   [DOI: 10.1021/acs.jpca.6c00238] [Supporting Data] [Supporting Code]
3. **Zakary, O.***; Lantto, P. Equivariant Neural Networks Reveal How Host–Guest Interactions Shape ^{129}Xe NMR in Porous Liquids. *J. Phys. Chem. Lett.* **2025**, 16, 12095–12103. (Selected as *Journal Supplementary Cover*)
   [DOI: 10.1021/acs.jpclett.5c02846] [Supporting Data] [Supporting Code] [Journal Cover]
4. **Zakary, O.***; Body, M.; Charpentier, T.; Sarou-Kanian, V.; Legein, C. Revealed Preferential Short-Range Anion Ordering in Disordered RbM_2O_5F (M= Nb, Ta) Pyrochlore-Type Oxyfluorides. *Inorg. Chem.* **2025**, 64, 5764–5777.
   [DOI: 10.1021/acs.inorgchem.5c00615] [Supporting Data] [Supporting Code]
5. **Zakary, O.***; Body, M.; Sarou-Kanian, V.; Arnaud, B.; Corbel, G.*; Legein, C. Different Magnitudes of Second-Order Jahn-Teller Effect in Isostructural $NaMO_2F_2$ (M= Nb, Ta) Oxyfluorides. *J. Alloys Compd.* **2025**, 1010, 177457.
   [DOI: 10.1016/j.jallcom.2024.177457] [Supporting Data] [Supporting Code]
6. **Zakary, O.***; Body, M.*; Charpentier, T.; Sarou-Kanian, V.; Legein, C. Structural Modeling of O/F Correlated Disorder in $TaOF_3$ and $NbOF_{3\text{-}x}(OH)_x$ by Coupling Solid-State NMR and DFT Calculations. *Inorg. Chem.* **2023**, 62, 16627–16640.
   [DOI: 10.1021/acs.inorgchem.3c02844] [Supporting Data]
7. **Zakary, O.***; Jacklin, T.; Lantto, P. Transport Properties in Carbon Nanotubes Unlocked by Machine Learning-Accelerated Molecular Dynamics and Noble Gas NMR Simulations. 2026, *Submitted for publication*.
8. **Zakary, O.***; Mailhiot, S. E.; Egleston, B. D.; Selent, A.; Kearsey, R. J.; Mareš, J.; Cooper, A. I.; Greenaway, R. L.; Telkki, V.-V.*; Lantto, P.* Unifying the Description of Type-II Porous Liquids Through Machine Learning-Accelerated Molecular Dynamics and ^{129}Xe NMR: Simulations Informing Experiments. 2026, *In preparation*.
9. **Zakary, O.*** A Universal Machine Learning-Driven Framework for Solving Disorder in Heteroanionic Materials. 2027, *In preparation*.
10. **Zakary, O.*** A Foundation Model for NMR Parameters Prediction Across Molecular and Solid-State Systems. 2027, *In preparation*.

***Software and Code Development*:**

1. **MACE_hyperparams** – This repository contains the methodology and tools for optimizing the hyperparameters of the Equivariant Message Passing Neural Network model MACE.
   [Repository: github.com/ozakary/mace_hyperparams]
2. **NMR-MatTen** – Python-based workflow for generating an NMR-ML model using the *E(3)-Equivariant Graph Neural Networks* architecture *MatTen*.
   [Repository: github.com/ozakary/NMR-MatTen]
3. **ClusterProbe** – Python tool for analyzing local environments around xenon atoms in molecular clusters and identifying anomalous structures.
   [Repository: github.com/ozakary/ClusterProbe]
4. **TrajSlicer** – Python tool for converting LAMMPS trajectories to XYZ format.
   [Repository: github.com/ozakary/TrajSlicer]
5. **xyz-proton-tracker** – Python tool for analyzing molecular dynamics trajectories in XYZ format,

specifically designed to identify and visualize different protonation states of oxygen atoms (OH^-, H_2O, H_3O^+) through color coding for OVITO visualization.
[Repository: github.com/ozakary/xyz-proton-tracker]

6. **xyz-dataset-splitter** – Python tool for splitting XYZ molecular dynamics trajectory files into training, validation, and test sets for machine learning applications.
   [Repository: github.com/ozakary/xyz-dataset-splitter]
7. **NMR-VASP** – VASP NMR parameter calculator that extracts and processes NMR data from OUTCAR files for material science applications.
   [Repository: github.com/ozakary/NMR-VASP]
8. **Lammps-Kokkos-Mace_Mahti-CSC** – This repository provides a recipe for the installation of LAMMPS with Kokkos GPU acceleration and MACE model for ML interatomic potential support on the Mahti CSC supercomputer.
   [Repository: github.com/ozakary/Lammps-Kokkos-Mace_Mahti-CSC]

***Artistic and Visual Outputs*:**

1. **Journal Supplementary Cover Design** – Designed supplementary cover for *The Journal of Physical Chemistry Letters*, Volume 16, Issue 46 (2025).
   [Published cover: pubs.acs.org/toc/jpclcd/16/46]

Research Supervision and Leadership Experience

***Supervision of Postgraduate Students*:**

Ph.D. student (1):

- Ossi Laurila (2025–present), NMR Research Unit, *University of Oulu*, Oulu, Finland. Co-supervised with Dr. Perttu Lantto. *Topic:* Machine learning-accelerated methods for investigating nuclear quantum effects.

M.Sc. students (5):

- Ossi Laurila (2024), NMR Research Unit, *University of Oulu*, Olulu, Finland. Co-supervised with Dr. Perttu Lantto. *Topic:* Machine learning-accelerated methods for investigating nuclear quantum effects.
- Rachid Belrhiti-Nejjar (2022), IMMM–UMR 6283 CNRS, *Le Mans Université*, Le Mans, France. Co-supervised with Prof. Jens Dittmer. *Topic:* Solid-state NMR of paramagnetic copper and nickel phenanthroline complexes.
- Samira Tepondjou (2022), IMMM–UMR 6283 CNRS, *Le Mans Université*, Le Mans, France. Co-supervised with Prof. Jens Dittmer. *Topic:* Characterization of hybrid perovskites by solid-state ^{207}Pb NMR.
- Odéric Claverie (2023), IMMM–UMR 6283 CNRS, *Le Mans Université*, Le Mans, France. Co-supervised with Prof. Jens Dittmer. *Topic:* NMR correlation experiments for the study of whole diatom cells enriched with ^{13}C and ^{15}N.
- Rim Hussein (2023), IMMM–UMR 6283 CNRS, *Le Mans Université*, Le Mans, France. Co-supervised with Prof. Jens Dittmer. *Topic:* NMR of microscopic-sized single crystals.

***Supervision of Undergraduate Students*:**
B.Sc. student (1):

- Matias Hintsanen (2025), NMR Research Unit, *University of Oulu*, Oulu, Finland. Co-supervised with Dr. Perttu Lantto. *Topic:* Equivariant neural networks for modeling noble gas interactions in molecular systems.

***Supervision Responsibilities*:** Research methodology development, progress monitoring and troubleshooting, manuscript preparation guidance, and training in computational chemistry.

Teaching merits

***Pedagogical Training*:** "Pédagogies de l'enseignement supérieur" (Pedagogies of Higher Education), *Le Mans Université*, Le Mans, France (2020).

***Teaching experience*:**

- **Computational Physics and Chemistry** (M.Sc./Ph.D. level, 16 h), Faculty of Science, *University of Oulu*, Oulu, Finland (2024–2025).
- **Simulation of Phenomena in Physical Sciences using Python** (B.Sc. level, 24 h), Faculté des Sciences et Techniques, *Le Mans Université*, Le Mans, France (2022–2023).
- **Metrology and Electricity** (B.Sc. level, 66 h), **Physical and Wave Optics** (B.Sc. level, 43 h), and **Learning and Evaluation Situation** (B.Sc. level, 6 h), IUT du Mans, *Le Mans Université*, Le Mans, France (2020–2022).

***Total teaching hours*:** 155 hours across five courses at B.Sc., M.Sc., and Ph.D. levels.

***Teaching material development*:** Developed open-source GUI software – *water-pimd-simulations*

Other key academic merits

- **2024–present**: Member of **follow-up groups** for Ph.D. students Anand Chekkottu Parambil (defended February 2026) and Piotr Sobiecki (expected defense 2028), *University of Oulu*, Oulu, Finland.
- **2025–present**: Member of the "*AI Group for Research*", guidelines on AI use in research, *University of Oulu*, Oulu, Finland.
- **2024–present**: Member of the "*Magnetic Resonance in Porous Media*" (MRPM) community.
- **2024–present**: Member of the "*Groupment AMPERE*" association.
- **2024–present**: Member of the "*Finnish Union of University Researchers and Teachers*".
- **2020–2023**: Member of "L'Association des Doctorants de l'Université du Mans", *Le Mans Université*, Le Mans, France. Promoting networking, interdisciplinary interaction, and scientific outreach among PhD students.

Scientific and societal impact

***Science communication and public engagement*:**

- **2023**: Organized and led laboratory visits and research demonstrations for high school students at IMMM–UMR 6283 CNRS, Le Mans University, Le Mans, France.
- **2020–2023**: Active participation in monthly science outreach events through ADoUM, including laboratory tours and Café-Thèse sessions for research dissemination to diverse audiences.